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UIDAI DATA HACKATHON REPORT

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Team ID

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Problem Statement

Statement: Unlocking Societal Trends in Aadhaar Enrolment and Updates

Identify meaningful patterns, trends, anomalies, or predictive indicators and translate them into clear insights or solution frameworks that can support informed decision-making and system improvements.

Who and how it impacts?

Operators / Service Providers

These include:

- Enrolment operators
- Authentication agencies
- Government service centers

Example:

Inefficiencies during Aadhaar Enrolment and updates result in increased retry rates and longer processing times because enrolment and authentication operators need to face frequent authentication failures without intelligent fallback mechanisms.

Case Study 1 : Authentication of Elderly Citizen

- Fingerprints worn out
- Multiple retries fail
- System does not switch automatically to:
 - OTP
 - Face authentication
 - Assisted verification

Problem: User is denied service even though identity is genuine - this leads to anomalies.

Overall, repeated authentication failures increase :

- System load
- Raise grievance volumes
- Pose reputational risks

Therefore, also impacting the trust and adoption of Aadhaar-based services across the ecosystem.



Approach

The proposed approach uses enrollment and biometric analytics to identify operational stress points that directly impact enrollment operators, authentication agencies and government service centers.

Enrollment dashboard captures both temporal and spatial patterns of enrollment activity, enabling detailed descriptive, comparative and trend analysis. It is analyzed to detect:

- Peak volumes
- Regional concentration
- Temporal surges like growth over month

The above contribute to operator overload and increased processing times.

Biometric dashboard reflects the periodic revalidation or correction of biometric details, especially for children transitioning into adulthood. This dataset further segments data by :

- Age Group
- District
- Weekday

These highlight scenarios where biometric capture failures are more likely, particularly among elderly and adult populations during high-load periods. By correlating peak enrolment windows with biometric intensity and demographic patterns, the approach enables early identification of high retry-risk environments.

These insights support the design of adaptive operational strategies such as:

- Workload redistribution – enforcing more workforce and operators to areas where enrollments are high.
- Context-aware authentication flows – High risk or normal region

Thereby, reducing service denial, prevents fake enrollments and controls misuse during peak enrollment drives while maintaining UIDAI security and compliance standards.



Datasets Used

This project utilises official aggregated datasets published by the Unique Identification Authority of India (UIDAI). The datasets provide comprehensive information on Aadhaar enrolment and biometric update activities across India and are suitable for large-scale demographic and geographic analysis.

1. Aadhaar Enrolment Dataset

Source: UIDAI Official Website

Description: This dataset provides aggregated information on Aadhaar enrolments across various demographic and geographic levels. It includes variables such as the date of enrollment, state, district, PIN code, and age-wise categories (0–5 years, 5–17 years, and 18 years and above).

Key Attributes Used:

- **date** – Enrolment date or reporting period
- **state** – State name
- **district** – District name
- **pincode** – PIN code of enrolment location
- **age_0_5** – Number of enrolments in the 0–5 years age group
- **age_5_17** – Number of enrolments in the 5–17 years age group
- **age_18_greater** – Number of enrolments for individuals aged 18 years and above

Purpose of Use:

- Analyse age-wise Aadhaar enrolment distribution.
- Identify geographic variations in enrolment at state, district, and PIN-code levels.

2. Aadhaar Biometric Update Dataset

Source: UIDAI Official Website

Description: This dataset contains aggregated information on biometric updates (modalities such as fingerprints, iris, and face). It reflects the periodic revalidation or correction of biometric details, especially for children transitioning into adulthood.

Key Attributes Used:

- **date** – Biometric update date or reporting period
- **state** – State name
- **district** – District name
- **pincode** – PIN code of update location
- **bio_age_5_17** – Number of biometric updates for individuals aged 5–17 years
- **bio_age_17_plus** – Number of biometric updates for individuals aged 17 years and above

Purpose of Use:

- Analyse biometric update trends across age groups.
- Compare biometric update volumes across regions.



Methodology

This project follows a structured analytics methodology using official UIDAI datasets and Power BI to analyse Aadhaar enrolment and biometric update patterns across demographic, geographic, and temporal dimensions. The methodology is aligned with the design and measures implemented in the final dashboards.

1. Data Collection

Two aggregated datasets were collected from the UIDAI official website:

1. Aadhaar Enrolment Dataset – capturing age-wise enrolments across states, districts, PIN codes, and time.
2. Aadhaar Biometric Update Dataset – capturing age-wise biometric updates across regions and dates.

Both datasets are anonymised and aggregated, ensuring privacy-compliant analysis.

2. Data Preprocessing and Cleaning

Before performing any analysis, the raw Aadhaar enrolment and biometric datasets were carefully preprocessed and cleaned to ensure accuracy. This step is crucial for eliminating errors and preparing the data for transformation and modelling.

1. Removal of Duplicate Records

- Identified and removed duplicate entries in both enrolment and biometric datasets to avoid double counting in totals and averages like showing different “peak district enrollments” for one state.

2. Standardising state and district names to avoid duplication

- Checked for missing or inconsistent values in State, District, and PIN code columns.
- Corrected inconsistencies to ensure uniformity, such as:
 - “Puducherry” instead of “Pondicherry”
 - “Odisha” instead of “Orissa”
 - “West Bengal” instead of “West Bangal”



3. Converting Date Columns to Proper Date Format

- All date fields were converted to standard Date/Time format to enable temporal analysis (Year, Month, Day).
- Resolved display issues such as dates appearing as ##### by correcting column widths and data formats in Excel and Power BI.

3. Data Transformation

Data transformation in this project was carried out using Power BI Power Query and DAX to prepare the raw UIDAI enrollment and biometric datasets for analysis. This step reshapes, aggregates, and enriches data to support dynamic and interactive dashboards.

1. Appending and Consolidating Tables – Multiple UIDAI enrollment and biometric files were appended into single fact tables since each table had different amount of data.

2. Derived Time Attributes – Extracted Year, Month, Month Name, Day Name, Day Type (Weekday/Weekend) from the date column.

3. Aggregation by Geographic Levels

- Enrollment and biometric data aggregated at:
 - State level
 - District level
 - PIN code level
- Facilitated drill-down analysis in dashboards.

4. State Mapping Table – Created this to merge queries that contained duplicate state names and transformed to one.

4. Data Modelling

1. Creation of Calculated Columns and Measures (DAX)

Enrolment Measures

- Total Enrollments = Sum of age_0_5, age_5_17, age_18_greater
- Average Daily Enrollments = Average enrolments per day
- Peak Daily Enrollments = Maximum daily enrolments
- Month-on-Month (MoM) Growth % = % change compared to previous month
- State-wise Average Enrollments = Average per state
- Peak District Enrollments = Maximum enrolments per district



Biometric Measures

- Total Biometrics = Sum of bio_age_5_17 + bio_age_17_plus
- Youth Biometrics = bio_age_5_17
- Adult Biometrics = bio_age_17_plus
- Total Districts Covered = Count of districts with updates
- Total PIN Codes Covered = Count of PIN codes with updates
- Highest Weekday = Weekday with maximum biometric activity

5. Analytical Approach

The analytical approach for this project focuses on extracting meaningful insights from Aadhaar enrolment and biometric update data through descriptive, comparative, and trend-based analysis. Using Power BI dashboards, the analysis explores demographic, geographic, and temporal patterns to support data-driven administrative decision-making.

1. Descriptive (Univariate) Analysis: Performed to understand the overall distribution and magnitude of enrolment and biometric updates.

Enrolment Dashboard:

- Total Aadhaar enrolments across the selected period
- MoM Growth %
- Average daily enrolments and peak enrolment days

Biometric Dashboard:

- Total biometric updates
- Distribution of biometric updates across age groups (5–17 and 17+)
- Total districts and PIN codes covered

2. Comparative (Bivariate) Analysis: Was used to compare two dimensions at a time, such as geography and time, or age group and enrolment volume.

Enrolment Dashboard:

- State-wise and district-wise comparison of enrolment volumes
- Month-wise enrolment trends to identify growth or decline
- Comparison of enrolment levels across age groups by region

Biometric Dashboard:

- Comparison of youth vs adult biometric updates across states
- District-wise comparison of biometric update volumes
- Comparison of biometric activity across weekdays and weekends



6. Data Visualisation

Data visualisation was implemented using Power BI to effectively present Aadhaar enrollment and biometric update insights through two interactive dashboards: Enrolment Dashboard and Biometric Dashboard.

Visualizations Used:

1. Enrolment Dashboard

This dashboard evaluates Aadhaar enrolment performance and growth patterns.

Visualizations Used:

- **KPI Cards:** Total enrolments, average daily enrolments, peak daily enrolments, state-wise averages, and Month-over-Month (MoM) growth percentage.
- **Line Charts:** Monthly enrolment trends and daily enrolment patterns.
- **Bar Charts:** Top states and districts by enrolment volume.
- **Donut Chart:** Age-wise enrolment distribution.
- **Tables:** Month-wise enrolment with MoM growth for performance monitoring.

2. Biometric Dashboard

This dashboard focuses on analyzing Aadhaar biometric updates across time, geography, and demographics.

Visualizations Used:

- **KPI Cards:** Total biometrics, youth biometrics, adult biometrics, total states, total districts, total pincodes, and highest weekday.
- **Line Charts:** Monthly biometric trends and age-wise biometric trends over time.
- **Bar & Column Charts:** State-wise, district-wise, and pincode-wise biometric activity; youth vs adult comparisons; weekday vs weekend analysis.
- **Donut Charts:** Proportional distribution of youth and adult biometrics.
- **Geographic Map:** Spatial distribution of biometric activity across regions.
- **Tables:** District-level and state-level summaries for detailed analysis.

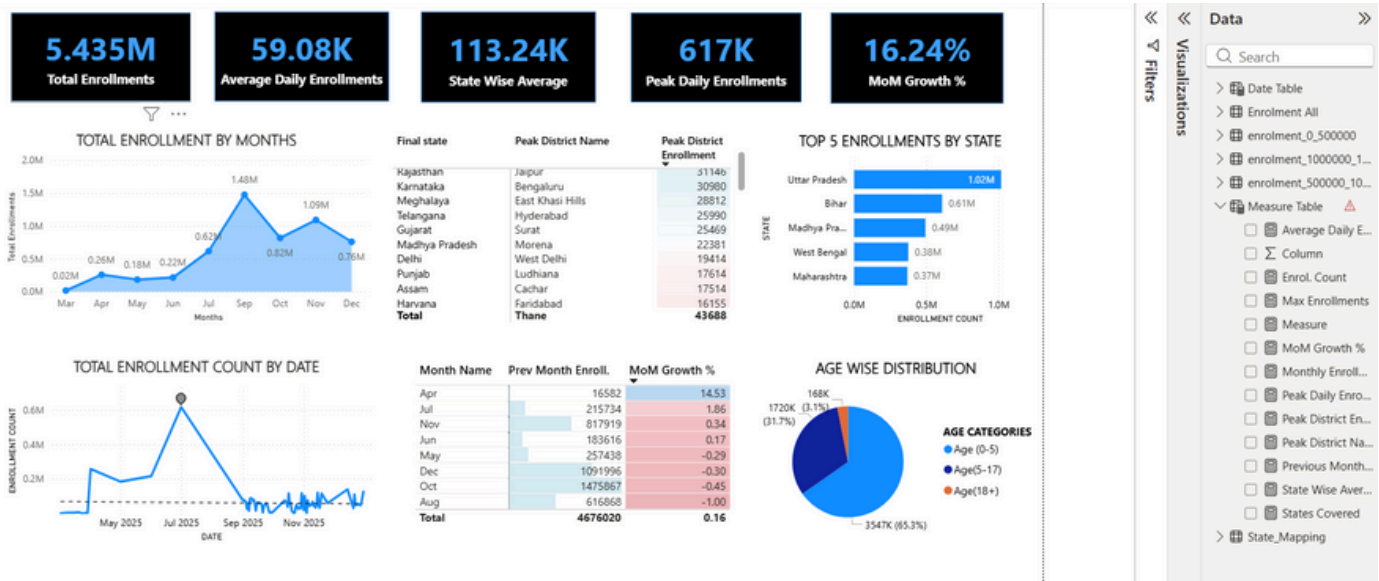
7. Tools Used

- **Microsoft Power BI:** Primary tool for data modeling, interactive dashboard creation, and visualization.
- **DAX (Data Analysis Expressions):** Used for advanced calculations such as growth metrics (DoD, WoW, MoM), KPIs, age-wise aggregation, and active vs dormant district classification.
- **Excel / CSV Files:** Used as source datasets for Aadhaar enrolment and biometric data.



Data Analysis & Visualisation

1) Aadhar Enrollment Data Dashboard



1) Key Performance Indicators (KPIs)

The dashboard highlights critical metrics including :

- Total Enrollments (5.435M)
- Average Daily Enrolments (59.08K)
- State-wise Average Enrolments (113.24K)
- Peak Daily Enrolments (617K)
- Month-on-Month (MoM) Growth (16.24%).

These KPIs offer a high-level summary of overall system workload, growth trends, and peak stress points within the enrolment ecosystem.

Measures for the above points:

1) Total Enrollments = SUMX('Enrolment All','Enrolment All'[age_0_5] + 'Enrolment All'[age_5_17] + 'Enrolment All'[age_18_greater])

2) Average Daily Enrollments = AVERAGEX(VALUES('Enrolment All'[date]),[Total Enrollments])

3) State Wise Average = AVERAGEX(VALUES('Enrolment All'[state]),[Total Enrollments])

4) Peak Daily Enrollments = MAXX(VALUES('Enrolment All'[date]),[Total Enrollments])

5) MoM Growth % = DIVIDE([Total Enrollments]-[Previous Month Enroll.],[Previous Month Enroll.])

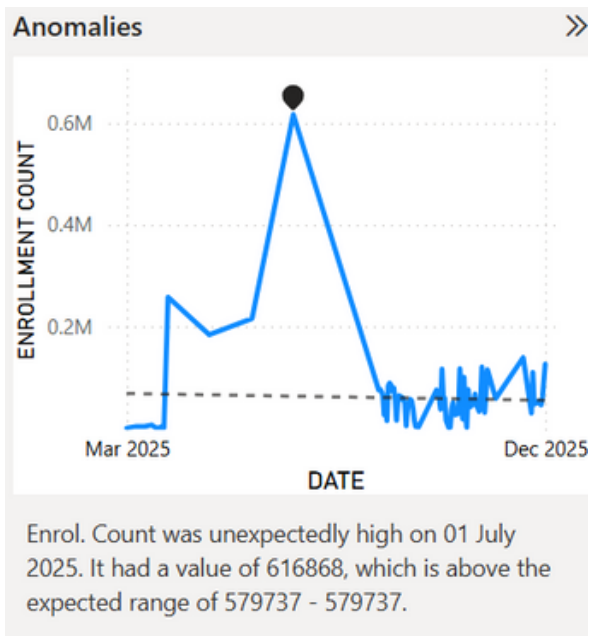


2) Temporal Analysis

- Monthly enrollment: reveals significant variation across the year, with a sharp increase observed in the month of September as 1.48 million (highest number of enrollments) and lowest in the month of March as 0.02 million - indicating periods of high and low operational demands.
- Daily enrollment: further highlights sudden spikes on specific dates, suggesting concentrated enrollment drives or service surges that can strain operator capacity and infrastructure if not anticipated.

Ex: Lowest enrollments (67) - 29th March, 2025

Highest enrollments(616868) - 1st July, 2025



Trend Line : Enrolment volumes show a steady upward growth, punctuated by sharp spikes during peak periods like enrollment drives or policy deadlines. These surges are followed by a normalization phase where activity stabilizes toward the long-term average, indicating that growth is driven by targeted initiatives rather than uniform daily distribution.

Anomaly: Highlights an unusually high spike in enrolment on 01 July 2025, where the enrollment count (363,639) significantly exceeded the expected range (342,888). This sudden deviation from the normal trend indicates an exceptional surge in enrolment activity.

3) Month-on-Month growth: Shown in a matrix format, it compares enrollment volumes across successive months, enabling the identification of growth accelerations and slowdowns. This insight is essential for forecasting demand and planning resources proactively.

Highest/positive growth can be seen from March to April (highlighted with dark shade of blue) ; whereas the lowest /negative growth can be seen from Sept to Oct(highlighted with dark shade of red).

At the end of the table, "Total" can also be seen that gives an aggregated growth percentage i.e. 16%



4) Geographic Distribution:

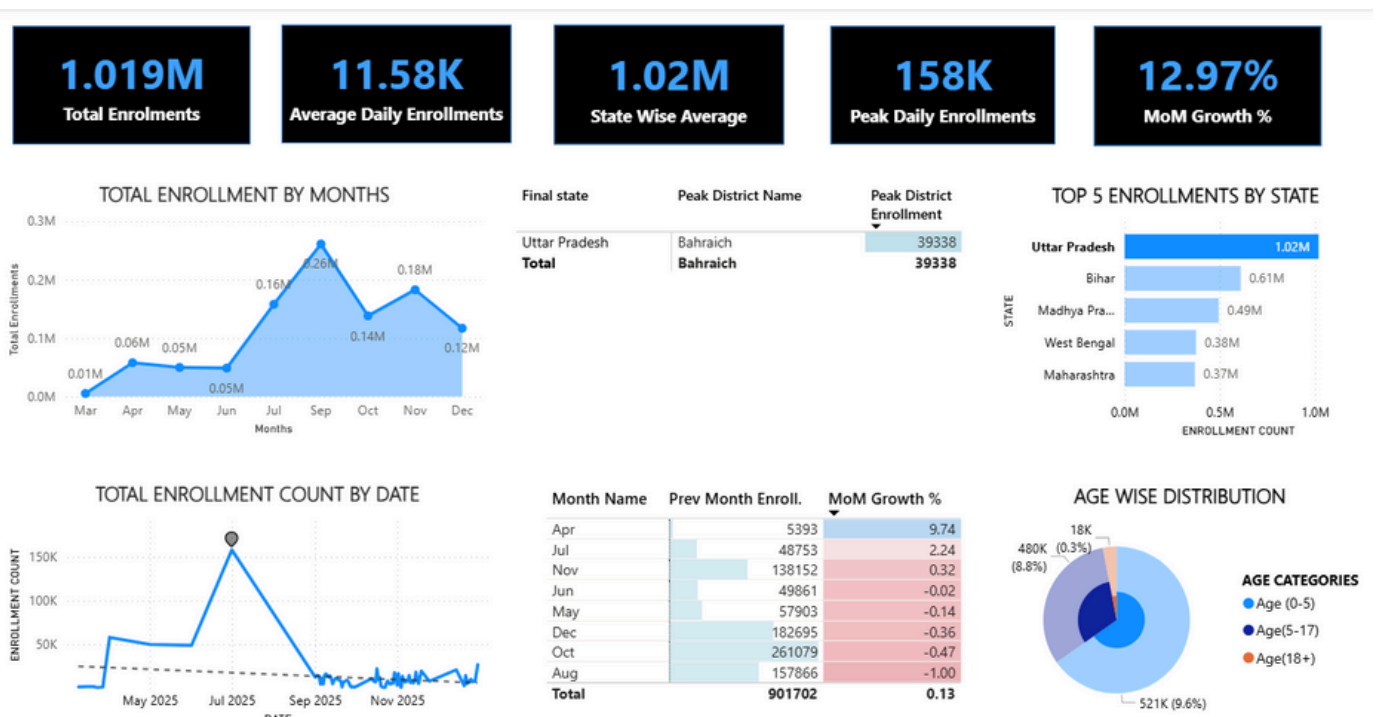
- State-wise analysis: Represented in a form of bar graph, it identifies regions with the top 5 enrollment volumes, with states such as Uttar Pradesh, Bihar, and Madhya Pradesh contributing a substantial share of total enrolments.
- District-level peak analysis: Visualized in table form, it shows all districts and their enrollments state-wise. It drills down to identify:
 - 1.High-burden districts(represented with darker shades of blue)
 - 2.Low-burden states(represented with darker shades of red).
 - 3.The states which have moderate enrollment values have been highlighted in white color.

Therefore, it enables targeted operational planning, staffing allocation, and infrastructure support where demand is concentrated.

5) Demographic Insights

Age-wise distribution shown in pie-chart shows that the majority of enrollments fall within the 0-5, 5-17 and 18+ age groups, indicating strong enrollment activity among children and youth populations. More than 50% children are engaged in this activity, therefore shows youth needs to participate more.

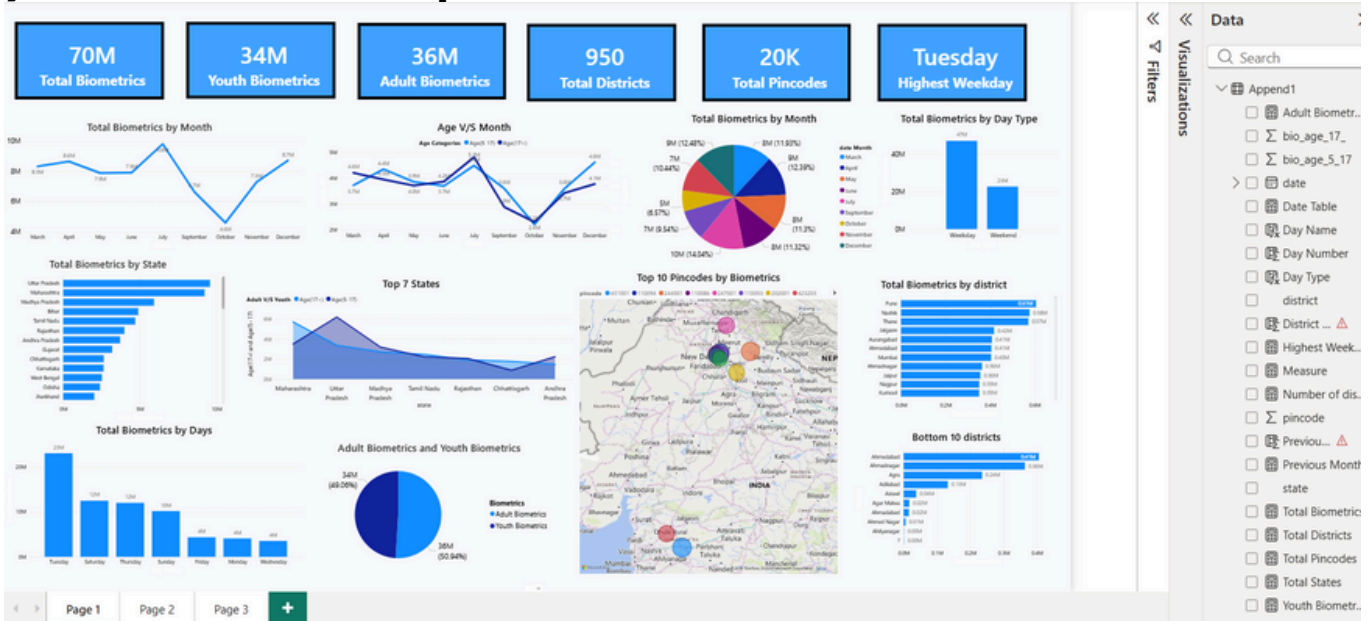
The graph below dynamically updates to other values. Here, **Uttar Pradesh** has been selected in "Top 5 Enrollments By State".





These insights are critical for planning biometric capture strategies suited to specific age categories. By integrating time-based trends, geographic concentration, and demographic segmentation, the dashboard supports workload management, peak-load preparedness, and efficient allocation of enrollment resources, ultimately enhancing the overall reliability of Aadhaar enrolment operations.

2) Aadhaar Biometric Update Dataset



1. KPI Overview

The KPI cards at the top of the dashboard present a consolidated national overview of Aadhaar biometric activity.

They display:

- **Total Biometrics** : 70 million biometric transactions
- **Youth Biometrics** : (5–17 years): 34 million
- **Adult Biometrics** : (18+ years): 36 million
- **Total Districts Covered** : 950
- **Total Pincodes Covered** : 20,000
- **Highest Weekday** : Tuesday

These KPIs provide an at-a-glance summary of the scale, demographic distribution, geographic coverage, and temporal behavior of biometric updates.

Analysis of KPI card :

The near-equal split between youth and adult biometrics indicates that Aadhaar biometric usage is driven by both youth and adults.

Coverage across 950 districts and 20,000 pincodes reflects deep penetration of Aadhaar services.

Identification of Tuesday as the highest activity weekday highlights a clear behavioral pattern.



Benefit to Government:

- These KPIs help the government quickly assess system reach and workload
- Knowing the highest weekday enables better planning
- Overall, this KPI summary acts as a real-time decision-support layer

2. Total Biometrics by Month

- The line chart displays the month-wise total Aadhaar biometric transactions across the year.
- Clear variations in activity levels are visible, with certain months showing higher biometric volumes than other
- The graph highlights peak months as well as low-activity months

Analysis:

- Biometric activity follows a seasonal trend rather than remaining constant throughout the year.
- Peak months indicate periods of increased citizen engagement, likely driven by administrative deadlines, welfare enrolments, migration, or educational cycles.
- Highest activity month is July and lowest is October

Benefit to Government:

- Helps the government understand when Aadhaar services are most used, so offices can be better prepared.
- Allows officials to avoid overcrowding and long waiting times for citizens.
- Enables better use of resources, reducing unnecessary operational costs.

3. Age-wise Biometrics vs Month

- The graph indicates that there was biometric activity among two age groups, namely youth (517 years) and adults (18+ years) in various months.
- Most months the adult biometrics are marginally higher than the youth biometrics.
- The two age groups have the same trend as they increase and decrease nearly on a monthly basis.
- The peak is observed during the middle of the year and a slight drop around October which was later replaced by a recovery during the later months.

Analysis

- The trend of the two age groups is similar, which indicates that the Aadhaar biometric activity is affected by shared external factors.
- The increased need is indicated by more adult activity.
- In some months, there is an increase in youth activity, perhaps associated with the admission to school.



- Decline of activity around October helps to suggest that the decreasing activity is a short-term phenomenon and not a permanent downturn.

Benefits to Government

- Enlightens the government on which age bracket accesses Aadhaar services most.
- Helps does not over crowd but prepares in advance of the peak months.
- Enables officials to enhance service delivery to both the youth and adults.
- Generates efficient operations of enrolment centres throughout the years

4. Total Biometrics by the Month (Pie Chart).

- The pie chart indicates the allocation of the total biometric activity in various months.
- Every slice is a percentage of the contribution of a month in the total of biometric count.
- Certain months will be having a higher percentage, whereas others will have a smaller percentage.

Analysis

- As can be seen in the chart, the biometric activity does not occur uniformly across the year.
- The percentage of the total activity is influenced by a few months, which implies the periods of high demand.
- Smaller shares in the months are indicators of low participation.

Benefits to Government:

- Assists the government to know the months that will be significant in working load.
- Enables improved scheduling of personnel and equipment when busy months occur.
- Helps in the smooth service delivery of Aadhaar at the peak times.

5. Weekday vs Weekend Biometrics

The bar chart compares total biometric transactions recorded on weekdays versus weekends. Weekdays show significantly higher biometric activity, while weekends contribute a smaller share of total transactions. This highlights a clear preference for Aadhaar services during regular working days.

Analysis:

The higher weekday activity suggests that most Aadhaar-related services are accessed during normal office hours, aligning with government office operations and citizen routines. Lower weekend activity may be due to limited service availability or reduced citizen engagement during non-working days. The contrast clearly shows how service demand varies based on the day type.



Benefits to Government:

This analysis helps the government optimize operational planning by ensuring adequate staffing and system capacity during weekdays. It also highlights opportunities to improve weekend accessibility through extended hours or special enrolment drives. Overall, it supports better workload management and enhances citizen convenience.

6. Total Biometrics by State

- The bar chart represents the total Aadhaar biometric transactions across different states.
- States such as Uttar Pradesh and Maharashtra record the highest biometric activity.
- Madhya Pradesh, Bihar, Tamil Nadu, and Rajasthan show moderate levels of biometric usage.
- Some states have comparatively lower biometric counts.

Analysis:

- States with higher biometric activity are likely influenced by larger populations and greater demand for Aadhaar-based services.
- Moderate and lower activity in other states may be due to differences in population size, awareness, or frequency of biometric updates.
- Overall, the variation clearly shows that Aadhaar biometric usage differs from state to state and is not evenly distributed across the country.

Benefits to Government:

- Helps the government identify states with higher service demand, where more enrolment centers and staff may be required.
- Supports better state-level planning for Aadhaar-related operations.
- Enables focused efforts in lower-activity states to improve accessibility and awareness of Aadhaar services.

7. Top 7 States – Adult vs Youth Biometrics

The area chart compares adult and youth biometric transactions across the top seven states. States like Uttar Pradesh and Maharashtra show the highest overall biometric activity for both age groups. In most states, adult biometrics are higher than youth biometrics, though the gap varies from state to state.

Analysis:

The dominance of adult biometrics reflects higher Aadhaar usage for employment, financial services, and government schemes in these states. Youth biometrics are also significant, particularly in highly populated states, indicating regular Aadhaar usage for education-related services. The gradual decline across states shows how biometric activity decreases as population size and service demand reduce.



Benefits to Government:

This comparison helps the government understand age-wise demand patterns at the state level. It supports targeted planning, such as focusing adult-related services in high-demand states while aligning youth enrolment drives with educational timelines. Overall, it enables better resource allocation, balanced service delivery, and more effective implementation of Aadhaar-based programs across states.

8. Top 10 Pincodes by Biometrics

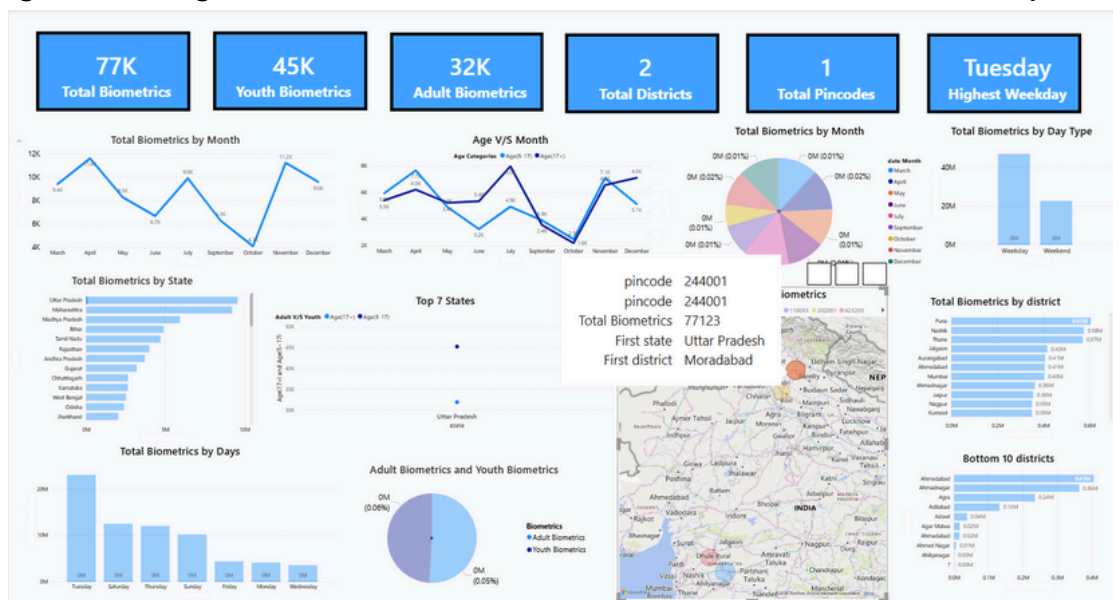
The map displays the top 10 pincodes based on Aadhaar biometric activity. Each location is represented by a circle, where larger circles indicate higher biometric density. A few pincodes clearly stand out with much bigger circles, showing significantly higher biometric transactions compared to others. Most of these high-density pincodes are located in urban or semi-urban regions.

Analysis:

The presence of larger circles in certain pincodes indicates high population concentration and frequent Aadhaar usage in those areas. These regions likely have better access to enrolment centers and higher demand for biometric updates due to employment, migration, and service requirements. Smaller circles represent pincodes with comparatively lower activity, highlighting uneven biometric distribution at the pincode level.

Benefits to Government:

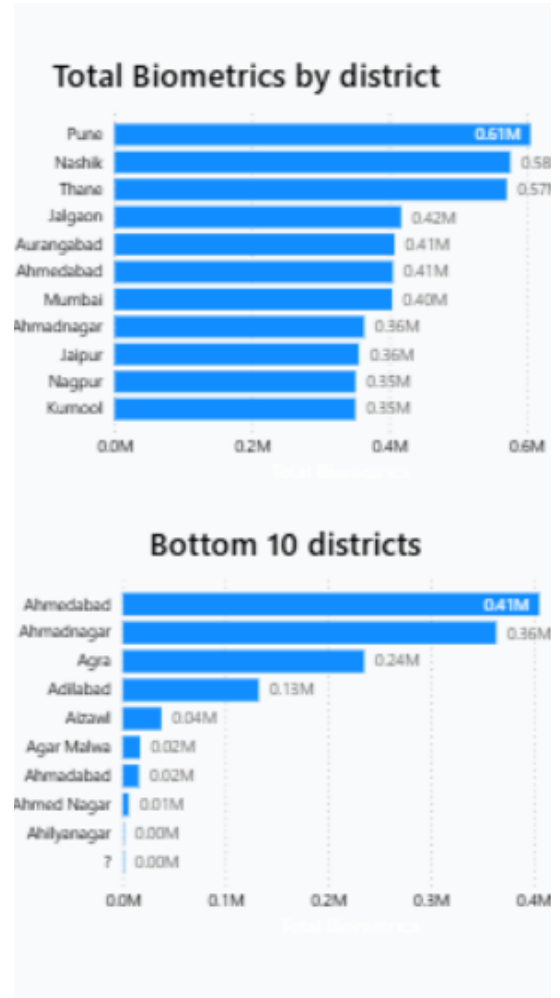
This visualization helps the government quickly identify biometric hotspots at a granular level. By focusing on high-density pincodes, authorities can ensure sufficient infrastructure and manpower where demand is highest. At the same time, nearby low-density areas can be targeted through mobile enrolment units and awareness campaigns, leading to balanced and efficient Aadhaar service delivery.





9.Total Biometrics by District

- The bar chart displays total Aadhaar biometric transactions across different districts. Districts such as Pune, Nashik, Thane, and Jalgaon show higher biometric activity, while others record moderate to lower volumes. This variation indicates that Aadhaar usage differs significantly across districts due to factors such as population size, urbanization, and administrative demand.
- This insight allows the government to prioritize high-demand districts for better staffing and service expansion, while also identifying lower-activity districts that may require focused attention. Overall, it supports balanced district-level planning and improves the efficiency of Aadhaar service delivery.



10. Lowest Biometrics by District

The bar chart highlights districts with the lowest Aadhaar biometric activity. Districts such as Ahmed Nagar, Bhoratabad, Agar Malwa, Aizawl, and Adilabad record significantly fewer biometric transactions compared to high-activity districts. The difference between the top and bottom districts is clearly visible.

Analysis:

Lower biometric activity in these districts may be influenced by smaller population size, lower frequency of Aadhaar updates, or limited access to enrolment centres. It may also indicate that a large portion of the population has already completed biometric enrolment, reducing the need for frequent updates. These districts represent low-demand regions rather than service inefficiency.

Benefits to Government:

Identifying low-activity districts helps the government plan targeted awareness campaigns and assess whether service accessibility needs improvement. It also allows authorities to optimize resource allocation, avoiding over-deployment in areas with limited demand. This ensures Aadhaar services remain cost-effective, balanced, and inclusive across all regions.



11.Total Biometrics by Day Name

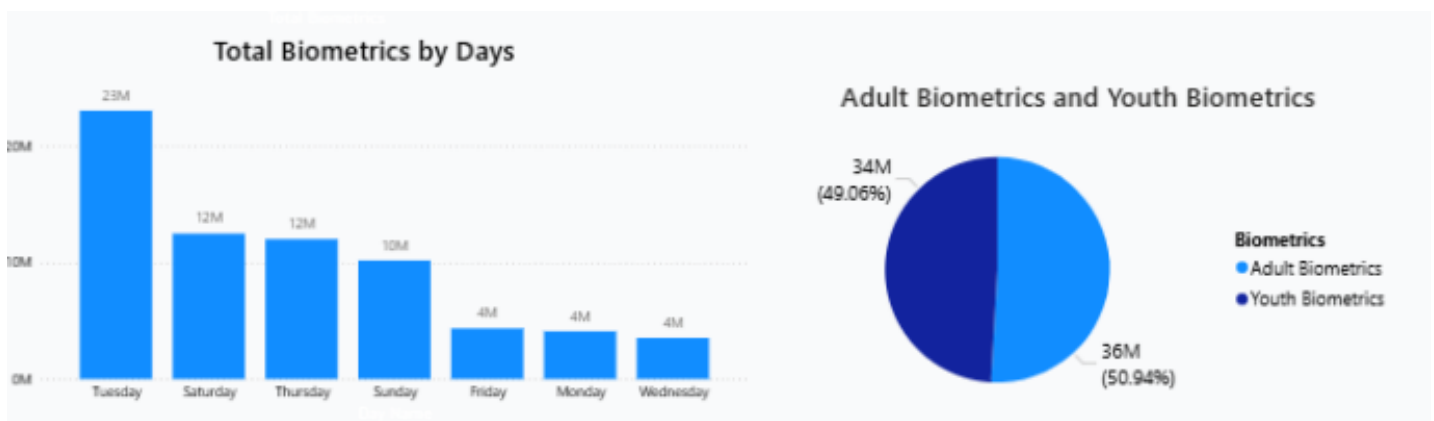
The bar chart presents the total Aadhaar biometric transactions recorded on each day of the week. Among all weekdays, Tuesday shows the highest biometric activity, followed by Saturday and Thursday. In contrast, days such as Monday and Wednesday record comparatively lower biometric volumes. This indicates that Aadhaar service usage varies significantly across different days of the week.

Analysis:

The uneven distribution of biometric activity across weekdays suggests that citizen behavior is influenced by work schedules, administrative routines, and personal convenience. Higher activity on Tuesday may be due to people preferring mid-week days to complete official tasks, while moderate activity on weekends indicates that some users rely on non-working days for Aadhaar updates. Lower activity on certain weekdays reflects predictable demand patterns rather than a lack of service availability.

Benefits to Government:

This analysis helps the government identify peak service days and plan operations accordingly. By deploying additional staff and technical support on high-activity days, authorities can reduce waiting times and avoid overcrowding at enrolment centers. On low-activity days, resources can be optimized or used for system maintenance and training. Overall, this insight supports efficient workforce planning, better service delivery, and improved citizen satisfaction in Aadhaar-related services.





12. Adult Biometrics vs Youth Biometrics

The pie chart shows the distribution of Aadhaar biometric transactions between adult and youth age groups. Adults account for a slightly higher share of total biometric activity, while youth biometrics form a significant but comparatively smaller portion. The overall distribution appears fairly balanced, with adults leading marginally.

Analysis:

The higher contribution from adults reflects the frequent need for Aadhaar updates related to employment, banking, government schemes, and identity verification. Youth biometric activity remains substantial, indicating regular usage during school admissions, exams, and welfare benefit enrolments. The close share between both groups suggests that Aadhaar services are relevant across age groups and used consistently throughout the population.

Benefits to Government:

This insight helps the government design age-specific service strategies, such as focusing adult updates on employment-related processes while planning youth-focused drives around academic cycles. It supports balanced resource planning and ensures Aadhaar services cater effectively to all age groups, improving overall service efficiency.